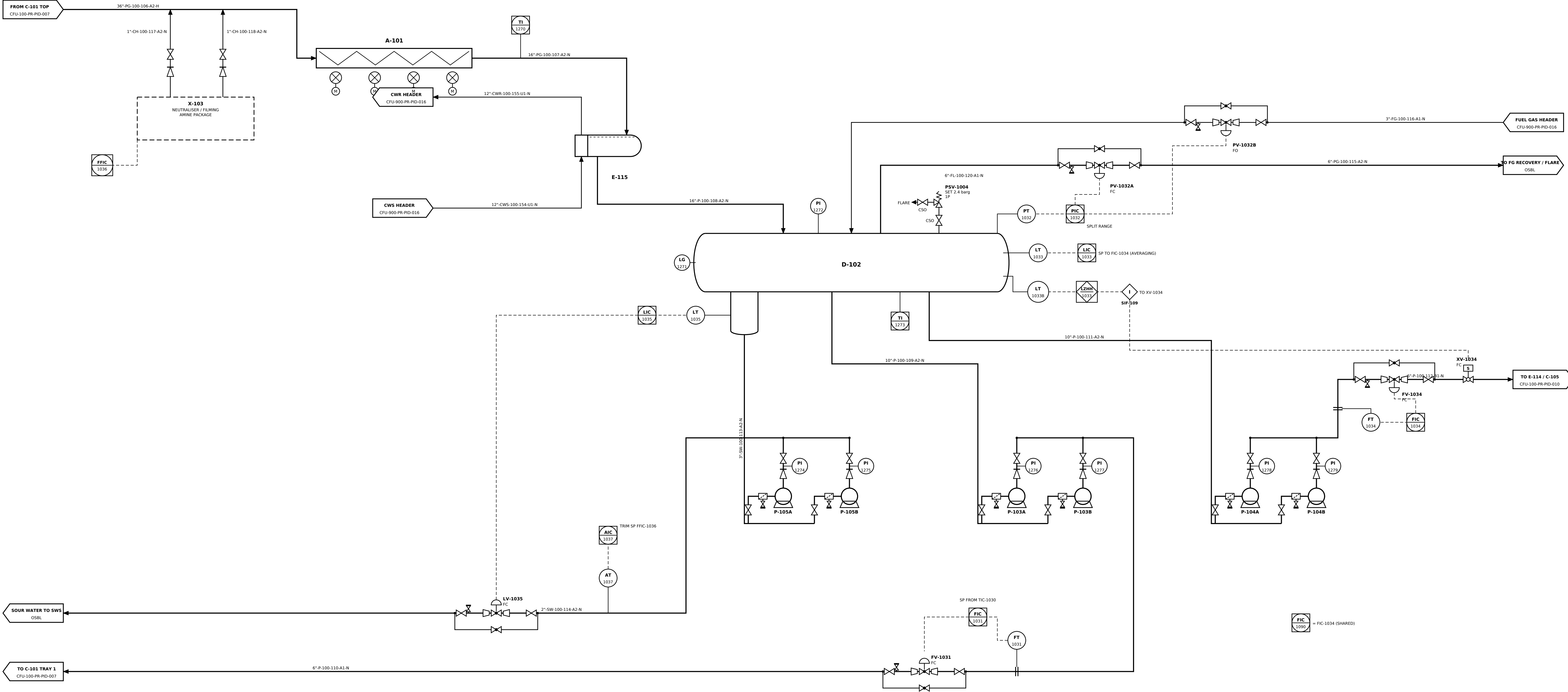


A-101 Atm. overhead condenser 1667 m2 bare; 2 bay(s) 6 m x 12 m, 4 fans DESIGN 9.7 barg / 140.5 C DUTY 33.75 MW; 4 x 90 kW fans MAT'L CS tubes, Al fins (A-101: CS + inlet ferrules, TI optional)	E-115 Atm. overhead trim condenser 740 m2; 2 shell(s) DESIGN 3.5 / 7 barg / 90 C; CA 3 mm DUTY 5.96 MW SHELL: OH vapour/condensate / TUBE: Cooling water MAT'L CS shell / TI tubes	D-102 Atm. overhead reflux drum (3-phase) ID 3.4 m x 12.0 m T/T; boot ID 1.5 m x 3.0 m DESIGN 3.5 barg / 140 C; CA 3 mm MAT'L CS (HIC-resistant); 6 mm CA	P-103A/B Atm. reflux RATED 159 m3/h @ 102 m, dP 7 bar MOTOR: 55 kW; API 682 Plan 11/52 DES. T 75 C; MAT'L S-6 (CS/L2Cr)	P-104A/B Unstabilised naphtha RATED 180 m3/h @ 232 m, dP 16 bar MOTOR: 160 kW; API 682 Plan 11/52 DES. T 75 C; MAT'L S-6 (CS/L2Cr)	P-105A/B Overhead sour water RATED 14 m3/h @ 51 m, dP 2 bar MOTOR: 5.5 kW; API 682 Plan 11/52 DES. T 75 C; MAT'L S-6 (CS/L2Cr)	X-103 Neutraliser / filming amine package (atm OH) Tank 2 m3 + 2 x 100 % metering pumps MOTOR: 0.75 kW MAT'L SS316
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- NOTES**
- Neutraliser and filming amine injected upstream of A-101; pH control AIC-1037 trims FFC-1036.
 - PIC-1032 split range: 0-50 % PV-1032B (FG make-up) closing, 50-100 % PV-1032A (off-gas) opening.
 - SIF-109 initiator / final element per SRS: LT-1033B high-high closes XV-1034 (see data issue log).
 - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
 - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
 - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST			
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING & INSTRUMENT DIAGRAM			
C-101 OVERHEAD SYSTEM & REFLUX DRUM D-102			
DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	10 OF 17
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-PR-PID-009			A